18.5 A Self-Calibrated Pipeline ADC with 200MHz IF-Sampling Frontend

Mikko Waltari, Lauri Sumanen, Tuomas Korhonen, Kari Halonen

Electronic Circuit Design Laboratory, Helsinki University of Technology, Finland

M. Waltari

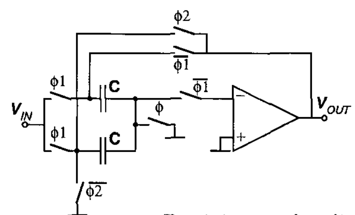
A 13b 50MSample/s pipeline ADC with digital self-calibration and IF-sampling frontend, using a 0.35µm BiCMOS process, achieves 76.5dB SFDR at 194MHz input. The chip occupies 6mm² and dissipates 715mW from a 2.9V supply.

See Digest page 314

Outline

- Introduction
- Prototype Architecture
- Front-End S/H
- Bootstrapped Switch
- Operational Amplifier
- Low Jitter Clock Buffering
- ADC Architecture
- Self-Calibration
- Experimental Results

Front-End



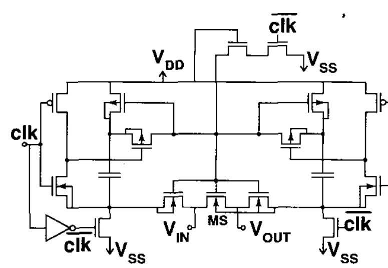
- Gain: 1 or 2
- 5 pF capacitors
- 3.8 Vpp diff. signal swing @2.9 V supply
- Bootstrapped switches; also the sampling switch
- High speed, high swing, low noise opamp

Introduction

Evolution toward software defined radio in cellular base stations ➔ need for high-speed, high-resolution ADCs with IF-sampling capability

Challenges:		Solutions:
Resolution	➔	Self-calibration
Sampling linearity	➔	Bootstrapped switch
Jitter	➔	Minimal on-chip clock buffering
Low voltage	➔	Rail-to-rail signal swing

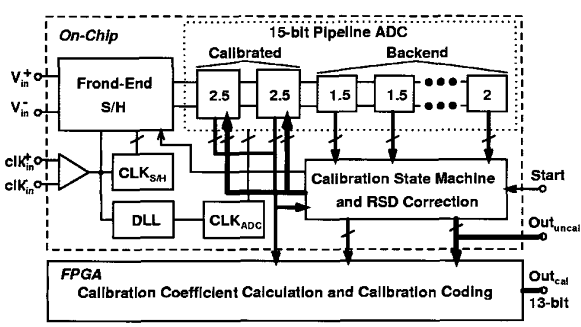
Double-Side Bootstrapped Switch



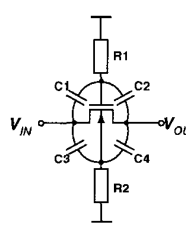
Gate voltage is average of V_{IN} and V_{OUT}

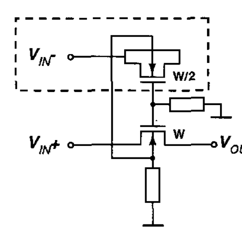
Bulk tracks the signal ➔ no bulk effect, no junction capacitances

Architecture



Feedthrough Cancellation

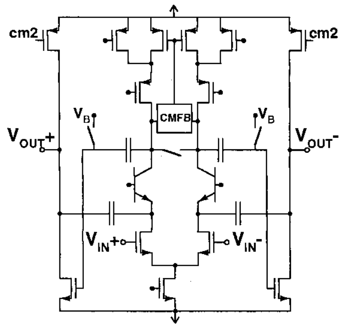




High-pass feedthrough paths past an open switch

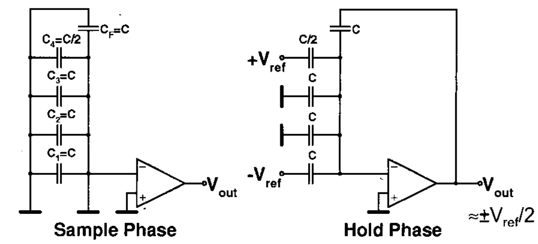
Feedthrough cancellation with a dummy switch

Operational Amplifier



- Low noise
- High speed
- >90 dB DC gain
- Capacitive level-shift between stages
- Cascode compensation

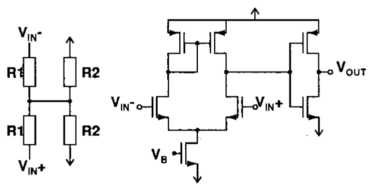
Measuring Capacitor Values



Extra C/2 capacitor added to MDAC

During measurement V_{out}≈±V_{ref}/2 ➔ opamp gain also calibrated

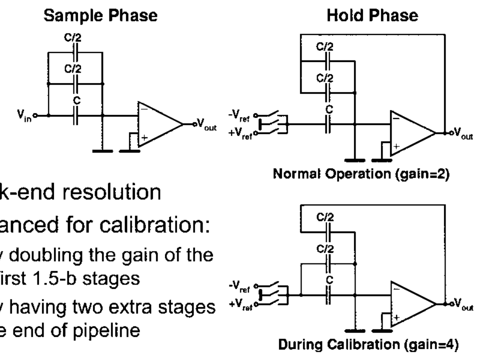
Sampling Clock Buffering



Input Clock Buffer

- Sampling clock doesn't go through clock generator ➔ low jitter
- ADC clock synchronized to sampling clock with DLL

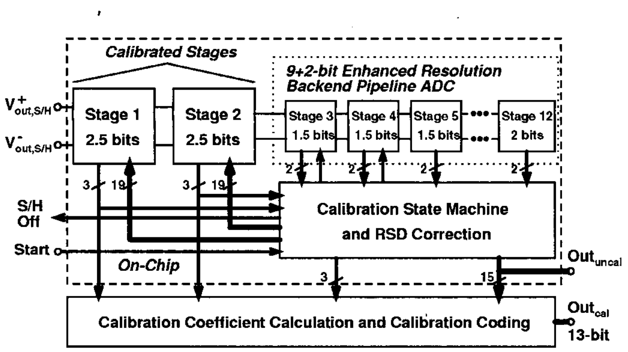
Using Backend in Calibration



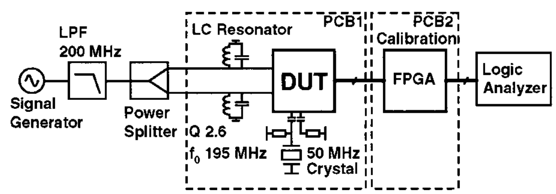
Back-end resolution enhanced for calibration:

1. By doubling the gain of the two first 1.5-b stages
2. By having two extra stages in the end of pipeline

ADC Architecture



Measurement Setup

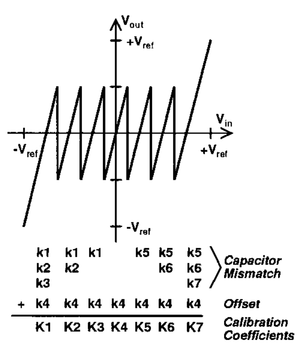


- Clock from 50 MHz crystal oscillator
- LC resonator decouples the current spikes produced by switched sampling capacitor
- Bit errors due to a timing error in the logic limited SNR to 55-60 dB, but didn't affect the linearity

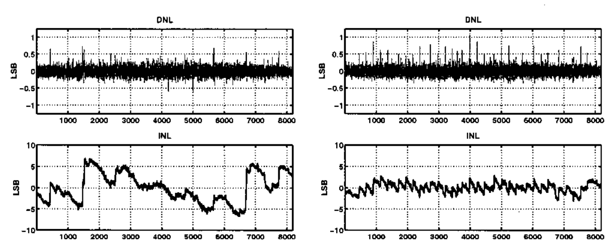
Calibration

Value of each unit capacitor is measured using the back-end stages

Calibration coefficients are calculated from the capacitor values



Static Linearity



Before calibration: INL=±7.1 LSB

After calibration: INL=±3.0 LSB

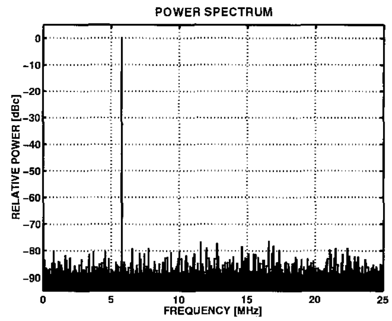
Continued on Page 493

Continued from page 251

Performance summary

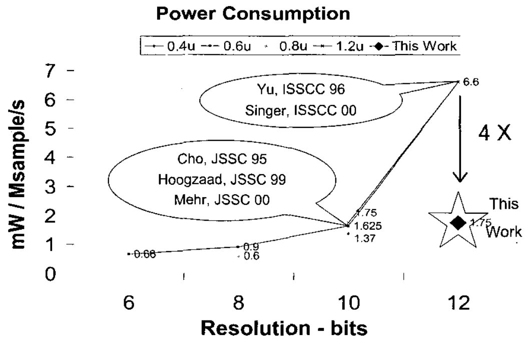
Technology	0.6u CMOS double poly process
Resolution	12 bit
Conversion rate	21 MSample/s
Supply Voltage	2.7 to 3.3 V
Power Consumption	35 mW at 3.0 V (Total ADC including error correction) 30 mW at 2.7 V
DNL	+0.5/-0.6
INL	+1.2/-1.0
Peak SNR	68.2 db at 10kHz signal input
Active die area	2.7x0.95 mm^2

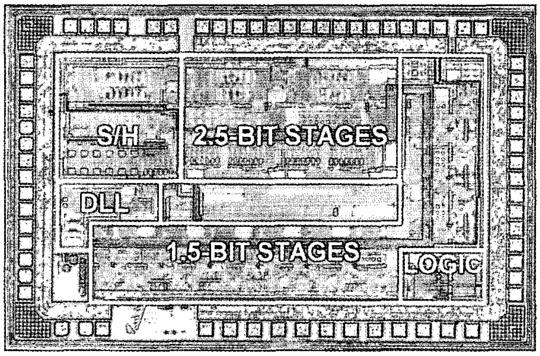
IF Spectrum



76.5 dB SFDR @ 194.2 MHz, -1 dBFS signal and 50 MHz clk

Benchmarking





Technology: 0.35-µm BiCMOS(SiGe) Area: 6.0mm²

Conclusion

Design ideas

- Interleaving for ADC architecture
- Split-Amp technique for Amp-sharing
- Comparator designed using
  - Charge averaging technique
  - Statistical simulation for minimum power

Design achievements

- Power is 4X lesser than any reported 12b ADC in any technology

Summary

Resolution	13 bits
Sampling Rate	50 MS/s
Input Range (diff.)	3.8 Vpp
Input Bandwidth	>200 MHz
DNL / INL (calibrated)	±1.0 / ±3.0 LSB
SFDR (@ 200 MHz)	76.5 dB
Power cons. @ 2.9 V	715 mW
Die Area	6.0 mm²
Technology	0.35-µm BiCMOS(SiGe)

Conclusions

IF-sampling self-calibrated pipeline ADC presented

3.8 Vpp signal swing from 2.9 V supply for maximizing signal-to-noise ratio

76.5 dB SFDR at 200 MHz IF